Extending Theorem 2 of *Robust Trust* to infinite M and Θ:
A three-tier theorem via the payoff-profile menu engine

Pedro Aldighieri

8 May 2026

Abstract

This note reports a three-tier infinite-dimensional extension of Theorem 2 ("Robustly Rationalizable Solution") from Dworczak & Smolin's *Robust Trust* (2026). The earlier restricted-game-plus-Lusin-lift route, which required three additional hypotheses (**A5-thick**, **A8c-attain**, **TRE-gen-Hall**) even for value optimality, has been replaced by a finite-dimensional *payoff-profile menu engine*. The reformulation is strictly stronger in three respects. (i) **Tier 1a** (existence of a value-optimal agent strategy plus ε-adversaries) holds under the paper's standing hypotheses alone; A5-thick and A8c-attain are no longer needed. (ii) **Tier 1b** (exact adversary attainment) is now isolated behind a clean *exact-contact* condition endogenous to the chosen optimal labeling. (iii) **Tier 2** (full robust rationalizability) replaces deterministic TRE-gen-Hall with a set-valued *menu-Hall* calibration, with the conclusion read q_β-almost-everywhere in the sense made precise below. A sharpness package (cone intersection lemma + no-free-dust theorem) shows menu-Hall is genuinely needed inside the engine: the obstruction is intrinsic multi-dimensional vector balance, uniform across all support patterns, and impervious to null-message dust under atomless τ. A separate classification result shows the witness's trust region is not primitive, so the witness exhibits a menu-engine artefact rather than a counterexample to the unrestricted infinite Theorem 2. A Tier 1a tightness theorem at the fixed-label level shows the ε-adversary of Tier 1a is the genuine ceiling whenever exact-contact fails on a τ-positive source set, and a companion cleanup theorem reduces the exact-contact hypothesis to its substantive content (contact nonemptiness) by absorbing the selector clause into Jankov–Von Neumann descriptive-set machinery. A further primitive strengthening shows that under atomless adviser-signal support τ alone, exact-contact holds for some value-optimal labeling — a τ-null Cantor canvas inside $\mathrm{supp}(\tau)$ supports a continuous surjection onto the optimal payoff menu, and repainting only this null set turns Tier 1b unconditional on the natural continuous-signal primitive.

1 The setting and the question

We use the notation of *Robust Trust*. Ω is finite with full-support prior μ_0; $s \in \Delta(\Omega)$ has state-conditional law $\pi(\cdot \mid \omega)$ and unconditional law $\tau = \sum_\omega \mu_0(\omega)\pi(\cdot \mid \omega)$; $M = \mathrm{supp}(\tau)$; $\theta \in \Theta$ (compact metric) is drawn from $f(\cdot \mid \omega)$, independently of s conditional on ω; A is compact metric; and $u(a, \omega, \theta)$ is bounded and continuous in a. Let Σ denote the agent's measurable strategies $\sigma : M \times \Theta \to \Delta(A)$ and B the misaligned adviser's measurable kernels $\beta : M \to \Delta(M)$. With probability α the adviser is aligned and reports s truthfully; with probability $1 - \alpha$ the adviser is misaligned.

The agent's robust objective is

$$U(\sigma) \;=\; \alpha\,\mathbb{E}_{\mathrm{id},\sigma}[u] + (1-\alpha)\,\inf_{\beta\in B} \mathbb{E}_{\beta,\sigma}[u], \qquad U^* \;=\; \sup_{\sigma\in\Sigma} U(\sigma), \tag{1}$$

and Definition 2 (*robust rationalizability*) asks for (σ^*, β^*) such that, at every on-path message m, the agent's prescribed continuation is Bayes-optimal under $P_{\beta^*}(\cdot \mid m)$. The paper's Theorem 2 establishes existence under the assumption that M and Θ are *finite*, via a finite Sion saddle. The question addressed here is whether existence can be extended when M and Θ are infinite.

The right reading of "for all m" in the infinite setting. Section 2 of *Robust Trust* states that, in infinite spaces, "statements involving 'for all' should be interpreted as 'for almost all' with respect to the underlying distributions." Definition 2 itself does not name the underlying distribution. The natural choice is the actual mixture message marginal under β^*,

$$q_{\beta^*} \;:=\; \alpha\,\tau \;+\; (1-\alpha)\int_M \beta^*(\cdot \mid s)\,\tau(ds),$$

because $P_{\beta^*}(\cdot \mid m)$ is itself a regular conditional probability defined q_{β^*}-a.e. Two consequences. First, the right reading of Definition 2 in the infinite setting is q_{β^*}-a.e., not literal-all and not merely τ-a.e. Second, the adversary is *not* required to be τ-absolutely continuous: β^* may place positive mass on a τ-null Borel set $N \subseteq M$, in which case $q_{\beta^*}(N) > 0$ and those messages are on-path for the mixture law. In the paper's finite proof, $\alpha > 0$ ensures every $m \in M$ has positive mixture mass, so all three readings coincide; in infinite M, τ-null but q_{β^*}-positive messages must satisfy the Bayes-optimality condition if the adversary uses them.

2 The engine

The earlier route worked in the agent's strategy space with the Balder weak topology on (Σ, T_λ), used Mertens's asymmetric minmax theorem on the restricted game, and invoked Lusin regularization to lift the restricted maximizer to the unrestricted game. That machinery is heavy and required three added hypotheses, the most consequential being A5-thick (a Lusin-thickness clause imposed on a representative of the maximizer).

The current route works instead in the finite-dimensional *payoff-profile space*

$$W \;\subseteq\; \mathbb{R}^{|\Omega|}, \qquad W \;:=\; \{w \in \mathbb{R}^{|\Omega|} : w(\omega) = \mathbb{E}_{\hat\sigma}[u(a,\omega,\theta) \mid \omega] \text{ for some } \hat\sigma \in \hat\Sigma\},$$

where $\hat\Sigma$ is the space of private strategies $\hat\sigma : \Theta \to \Delta(A)$. W is a compact convex subset of $\mathbb{R}^{|\Omega|}$. The agent's optimization reduces to choosing a Borel labeling $w : M \to W$ (a payoff-profile strategy); the misaligned adviser optimizes over Borel kernels $\beta : M \to \Delta(M)$; the value of the game depends on w only through the compact *menu* $C := \overline{w(M)} \subseteq W$.

This finite-dimensional reformulation has three advantages. First, the space $\mathcal{K}(W)$ of nonempty compact subsets of W is itself compact under the Hausdorff metric, so existence of an optimal menu C^* comes for free without any thickness or representative-choice hypothesis. Second, a closure-pruning step cleanly identifies the support $C^\dagger := \overline{w^*(M)} \subseteq C^*$ of an optimal labeling, decoupling existence from contact-set geometry. Third, the rowwise contact set

$$G(s) \;:=\; \{m \in M : s \cdot w^*(m) = \min_{z \in C^\dagger} s \cdot z\}$$

becomes a finite-dimensional object whose existence and selectability properties are easy to state and easy to fail in tight examples.

3 Main theorem

The standing hypotheses of *Robust Trust* are kept verbatim (Ω finite, μ_0 full support, A, Θ compact metric, u bounded continuous in a, conditional independence of s, θ given ω, all measurability Borel). The new theorem has three tiers, with the second and third each layering one additional endogenous condition.

Tier 1a (standing hypotheses alone). There exists $\sigma^* \in \Sigma$ with $U(\sigma^*) = U^*$. For every $\varepsilon > 0$, there exists $\beta_\varepsilon \in B$ with $U(\beta_\varepsilon, \sigma^*) \leq U^* + \varepsilon$.

Exact-contact condition. With w^* the chosen optimal labeling and $C^\dagger := \overline{w^*(M)}$, the rowwise contact set $G(s)$ is nonempty for τ-a.e. s and $s \mapsto G(s)$ admits a Borel measurable selector $m^* : M \to M$ with $m^*(s) \in G(s)$ τ-a.e.

Tier 1b (+ exact-contact). With $\beta^*(dm \mid s) := \delta_{m^*(s)}(dm)$, $U(\beta^*, \sigma^*) = \inf_{\beta \in B} U(\beta, \sigma^*) = U^*$.

Menu-Hall condition. There exists a set-valued kernel $\kappa : M \to \Delta(M)$ supported on $G(s)$ (i.e. $\kappa(G(s) \mid s) = 1$ for τ-a.e. s) such that, for the joint coupling $\gamma_\alpha := \alpha\,(\mathrm{id}, \mathrm{id})_{\#}\tau + (1-\alpha)\,\tau \otimes \kappa$ on $M \times M$ with second-marginal $q := (\gamma_\alpha)_2$, the posterior satisfies $P_{\gamma_\alpha}(\cdot \mid m) \in B(m)$ for q-a.e. m, where $B(m) := \{\mu \in \Delta(\Omega) : \hat\sigma^*(m) \in \arg\max_{\hat\sigma'} U(\hat\sigma', \mu)\}$ is the Bayes cone for the agent's continuation at message m.

Tier 2 (+ menu-Hall). With $\beta^* := \kappa$, $\hat\sigma^*(m) \in \arg\max_{\hat\sigma'} U(\hat\sigma', P_{\gamma_\alpha}(\cdot \mid m))$ for q-a.e. m. Hence σ^* is robustly rationalizable in the paper's a.e./on-path sense. When $\alpha > 0$, $q \geq \alpha\tau$, so the conclusion also holds τ-a.e.

4 Proof strategy in one paragraph

Tier 1a. The value of the game depends on w only through the menu C, and the menu-value functional $F(C) := \int_M [\alpha \max_{w \in C} m \cdot w + (1-\alpha) \min_{w \in C} s \cdot w]\,\tau(ds)$ is upper semicontinuous on the Hausdorff-compact $\mathcal{K}(W)$. Hence an optimal C^* exists, an optimal labeling $w^* : M \to C^*$ realizing $\max_{w \in C^*} m \cdot w$ pointwise exists by Aliprantis–Border 18.19, and a profile-realization sub-lemma (KRN selection on $\hat\Sigma \twoheadrightarrow W$) lifts w^* to a private strategy $\hat\sigma^*$. The closure-pruning step $C^\dagger := \overline{w^*(M)}$ preserves the value because $\min_{w \in C^\dagger} s \cdot w \geq \min_{w \in C^*} s \cdot w$ pointwise and C^* was already optimal. The ε-adversary is built by Jankov–von Neumann selection on the ε-thick approximate contact set $G_\varepsilon(s) := \{m : s \cdot w^*(m) \leq \min_{C^\dagger} s \cdot w + \varepsilon\}$.

Tier 1b. Under exact-contact, replace G_ε by G and the selector m^* delivers $\beta^*(\cdot \mid s) := \delta_{m^*(s)}$ exactly attaining the rowwise minimum τ-a.e.

Tier 2. Under menu-Hall, the support-function form $\int_E \phi(P_{\gamma_\alpha}(\cdot \mid m))q(dm) \leq \int_E h_{B(m)}(\phi)q(dm)$ for every measurable E and continuous affine ϕ is equivalent (by countable separation in finite-dim $\Delta(\Omega)$) to pointwise membership $P_{\gamma_\alpha}(\cdot \mid m) \in B(m)$ q-a.e., which is exactly the Bayes-optimality requirement of Definition 2. Since $q \geq \alpha\tau$ when $\alpha > 0$, q-a.e. upgrades to τ-a.e.

5 Tier 1a: from menu existence to ε-adversaries

Lemma 1 (L1 — menu-value equivalence). *For any Borel labeling $w : M \to W$ with menu $C := \overline{w(M)}$,*

$$U(\sigma_w) \ = \ \alpha \int_M m \cdot w(m)\, \tau(dm) + (1-\alpha) \int_M \min_{z \in C} s \cdot z\, \tau(ds).$$

The first term equals $\int_M \max_{z \in C} m \cdot z\, \tau(dm)$ at any labeling that selects pointwise maximizers on C.

Sketch. The aligned term integrates the agent's payoff at the truthful posterior m against the chosen profile $w(m)$; the misaligned term integrates the rowwise infimum over messages, which equals the infimum over $w(M)$ of $s \cdot w$, which by continuity equals the minimum over $\overline{w(M)}$. □

Lemma 2 (L2 — existence of an optimal menu). *There exists $C^* \in \mathcal{K}(W)$ maximizing $F(C) := \int [\alpha \max_{w \in C} m \cdot w + (1-\alpha) \min_{w \in C} s \cdot w] \tau(ds)$.*

Sketch. $\mathcal{K}(W)$ is compact under Hausdorff distance because W is compact in $\mathbb{R}^{|\Omega|}$. The aligned term is continuous in C (maximum of a continuous function over a Hausdorff-continuous compact set), and the misaligned term is upper semicontinuous in C (minimum is u.s.c. in C). Hence F is u.s.c., and $\mathcal{K}(W)$-compactness gives a maximizer. □

Lemma 3 (L3 — closure-pruning). *Let $w^* : M \to C^*$ be an optimal Borel labeling and $C^\dagger := \overline{w^*(M)}$. Then $F(C^\dagger) = F(C^*) = U^*$.*

Sketch. $w^*(m) \in C^\dagger \subseteq C^*$, so the aligned term is unchanged pointwise. $C^\dagger \subseteq C^*$ implies $\min_{C^\dagger} s \cdot w \geq \min_{C^*} s \cdot w$, so the misaligned term weakly improves. Optimality of C^* forces equality. □

Lemma 4 (L4 — ε-adversary realization). *For every $\varepsilon > 0$, define $G_\varepsilon(s) := \{m \in M : s \cdot w^*(m) \leq \min_{z \in C^\dagger} s \cdot z + \varepsilon\}$. Then $s \mapsto G_\varepsilon(s)$ has analytic graph and admits a universally measurable selector m^*_ε. The kernel $\beta_\varepsilon(\cdot \mid s) := \delta_{m^*_\varepsilon(s)}$ satisfies $U(\beta_\varepsilon, \sigma^*) \leq U^* + (1-\alpha)\varepsilon$.*

Sketch. $G_\varepsilon(s)$ is the preimage under $m \mapsto s \cdot w^*(m)$ of an interval of width ε shifted by an essentially-bounded measurable function $s \mapsto \min_{C^\dagger} s \cdot w$; its graph is analytic. Jankov–von Neumann [2, Thm. 18.13] provides a universally measurable selector. The payoff bound follows by plugging β_ε into the misaligned term of U. □

Tier 1a follows: existence of σ^* by Lemmas 2 and 3 plus the profile-realization sub-lemma; the ε-adversary by Lemma 4.

6 Tier 1b: exact-contact

The exact-contact condition isolates the gap between ε-attainment and exact attainment. There are several sufficient routes, all standard:

- **Closed image.** If $w^*(M) \subseteq W$ is closed, then $C^\dagger = w^*(M)$, the rowwise minimum on $C^\dagger$ is attained on $w^*(M)$, and any minimizer pulls back to $G(s) \neq \emptyset$.
- **Closed-graph correspondence.** If $\Gamma(s) := \arg\min_{w \in C^\dagger} s \cdot w$ is upper hemicontinuous with closed values, the measurable maximum theorem [2, Thm. 18.19] delivers a Borel measurable selector.

- **u.h.c. Bayes-action.** If the agent's Bayes-action correspondence at message m is u.h.c. with closed compact values, then choosing $\hat{\sigma}^*$ as a continuous selector yields a closed labeling, hence closed image.

Under exact-contact, Tier 1b is immediate: $\beta^*(\cdot \mid s) := \delta_{m^*(s)}$ attains $\min_{w \in C^\dagger} s \cdot w$ pointwise τ-a.e., so $U(\beta^*, \sigma^*) = U^*$.

7 Tier 1a tightness and Tier 1b cleanup

Two structural facts complete the Tier 1a/1b picture at the fixed optimal σ^*. Theorem 5 shows that the ε-adversary of Lemma 4 is the genuine ceiling whenever the rowwise contact set $G(s)$ is empty on a τ-positive source set; Theorem 7 shows that the "Borel selector" clause of the exact-contact condition (§6) is automatic once contact nonemptiness is granted. Together they classify exact-contact as a single, irreducible bridge between Tier 1a and Tier 1b (Corollary 8).

Throughout, write $h(s,m) := s \cdot w^*(m)$ and $r(s) := \min_{z \in C^\dagger} s \cdot z = \inf_{m \in M} h(s,m)$ (last equality by Lemma 3); h is bounded Borel, r is continuous (lower envelope of a compact family of linear forms over compact $C^\dagger$).

Theorem 5 (Fixed-label tightness)**.** *Let $E := \{s \in M : G(s) = \emptyset\}$. The set E is coanalytic, hence universally measurable; suppose $\tau(E) > 0$ in the τ-completion. Then:*

(i) For every $\varepsilon > 0$ there is $\beta_\varepsilon \in B$ with $\int\int h(s,m)\,\beta_\varepsilon(dm \mid s)\,\tau(ds) \le \int r\,d\tau + \varepsilon$.

(ii) No $\beta \in B$ satisfies $\int\int h(s,m)\,\beta(dm \mid s)\,\tau(ds) = \int r\,d\tau$.

Proof. The graph $\Gamma := \{(s,m) \in M \times M : h(s,m) = r(s)\}$ is Borel (zero set of the bounded Borel function $f(s,m) := h(s,m) - r(s)$ on the standard Borel space $M \times M$). Its s-projection $E^c = \pi_M(\Gamma)$ is analytic; E is coanalytic; both are universally measurable.

(ii). Suppose some $\beta \in B$ attains the infimum. With $\nu(ds, dm) := \tau(ds)\,\beta(dm \mid s)$, $0 = \int f\,d\nu$ and $f \ge 0$ Borel, so $f = 0$ ν-a.e., i.e. $\int_M \beta(G(s)^c \mid s)\,\tau(ds) = 0$. The integrand is nonnegative Borel, so $\beta(G(s) \mid s) = 1$ for τ-a.e. s. On E this forces $\beta(\emptyset \mid s) = 1$, contradicting $\tau(E) > 0$.

(i). $G_\varepsilon := \{f \le \varepsilon\}$ is Borel with nonempty sections by definition of inf. M compact metric is standard Borel, so Jankov–Von Neumann [2, Thm. 18.13] yields a universally measurable selector m^u_ε. Since τ is a probability on a standard Borel space, m^u_ε admits a Borel modification $\tilde{m}_\varepsilon$ equal τ-a.e. (see [3]). Set $\beta_\varepsilon(\cdot \mid s) := \delta_{\tilde{m}_\varepsilon(s)}$. Then $\int h(s, \tilde{m}_\varepsilon(s))\,\tau(ds) \le \int (r(s)+\varepsilon)\,\tau(ds) = \int r\,d\tau + \varepsilon$. □

Remark 6 (Dust does not repair the obstruction)*.* The q_β-a.e. reading of Definition 2 (§1) allows β to place positive mass on τ-null messages, which affects the per-message Bayes-optimality requirement but *not* the rowwise equality $h(s,m) = r(s)$ used here. Routing β-mass to a τ-null dust point cannot create a rowwise minimizer for a source $s \in E$ if no message in M satisfies the equality. Hence Theorem 5 is robust to the q_β-a.e. reading.

Theorem 7 (Selector is automatic)**.** *Under the standing hypotheses, with M standard Borel and w^* Borel, the following are equivalent:*

(a) (exact-contact, §6) $G(s) \neq \emptyset$ τ-a.e., and there exists a Borel selector $m^* : M \to M$ with $m^*(s) \in G(s)$ τ-a.e.;

(b) (cleanup) $G(s) \neq \emptyset$ τ-a.e.

Sketch. (a)⇒(b) trivial. (b)⇒(a): Γ is Borel with τ-a.e. nonempty sections; Jankov–Von Neumann gives a universally measurable selector on $P := \{s : G(s) \neq \emptyset\}$; the standard τ-a.e. Borel-modification lemma on a standard Borel space supplies a Borel selector. Extend by any Borel default on $M \setminus P$ (a τ-null set). □

Theorem 7 reduces the three sufficient routes of §6 (closed image, closed-graph correspondence, u.h.c. Bayes-action) to a single substantive content: each acts through contact nonemptiness, while the "Borel selector" half is descriptive-set machinery rather than a primitive structural hypothesis.

Corollary 8 (The bridge is one condition)**.** *For the chosen optimal σ^*, with w^* Borel and M standard Borel,*

$$\exists\, \beta^* \in B : \iint h(s,m)\, \beta^*(dm \mid s)\, \tau(ds) = \int r\, d\tau \iff G(s) \neq \emptyset \text{ for } \tau\text{-a.e. } s.$$

Sketch. (⇒) Theorem 5 (ii) contrapositive. (⇐) Theorem 7 provides a Borel $m^*(s) \in G(s)$ τ-a.e.; then $\beta^*(\cdot \mid s) := \delta_{m^*(s)}$ attains the infimum. □

What this does *not* establish. Theorem 5 is fixed-label: it concerns a specific optimal σ^* and its labeling w^*. A *primitive* obstruction — showing that for some primitive $(\Omega, \mu_0, \tau, A, \Theta, u, \alpha)$ *every* value-optimal σ^* has τ-positive contact failure — is not proved here and remains the right open question for certifying Tier 1a's ceiling as primitively sharp. Tier 1a's value-side conclusion and Tier 2's menu-Hall requirement are unaffected. The deletion-compatible Hall duality theorem flagged in the project record remains the right open question for Tier 2.

What *not* to substitute for exact-contact. τ-a.e. continuity of w^* is the wrong regularity notion: a single τ-null discontinuity can be the rowwise minimizer for a τ-positive source set, and the adversary may carry positive q_β-mass to it. Strict concavity in the agent's action plus a.e. unique-argmax hypotheses do not, by themselves, bypass exact-contact; only w^* continuous on all of M (not just τ-a.e.) reduces exact-contact to a free consequence.

8 Tier 1b strengthening: unconditional under atomless τ

§6 and §7 characterised exact-contact as the irreducible bridge between Tier 1a's ε-ceiling and Tier 1b's exact attainment, for a fixed optimal labeling w^*. This section delivers a *primitive* strengthening: under the standing hypotheses augmented by atomlessness of τ alone, exact-contact holds for *some* value-optimal labeling, and Tier 1b becomes unconditional.

Theorem 9 (Tier 1b under atomless τ)**.** *Assume the standing hypotheses of §1 and τ atomless on $M = \operatorname{supp}(\tau)$. Then there exists a value-optimal agent strategy σ^* whose payoff-profile labeling $w^* : M \to W$ satisfies $G(s) \neq \emptyset$ for every $s \in M$. Consequently, by Theorem 7, an exact adversary $\beta^* \in B$ exists with $U(\beta^*, \sigma^*) = U^*$.*

Proof. Apply Lemmas 1–4 to obtain an optimal Borel labeling $w_0 : M \to W$ with closure-pruned support $C_0^\dagger := \overline{w_0(M)}$ compact in W, $F(C_0^\dagger) = U^*$.

Null Cantor subset lemma. Because τ is atomless and $M = \operatorname{supp}(\tau)$, M contains no isolated points. We construct N by a Cantor scheme. Choose two distinct points $m^0, m^1 \in M$; by atomlessness and inner-regularity of τ on the Polish space M [3, Thm. 7.1.7, Vol. II], pick disjoint compact neighborhoods F^0, F^1 of m^0, m^1 with diameters $< 1/2$ and $\tau(F^0 \cup F^1) < 1/2$. Iterate: given F^t for $t \in \{0,1\}^n$, since F^t has no isolated points and τ is atomless, pick two distinct interior points m^{t0}, m^{t1} and disjoint compact neighborhoods $F^{t0}, F^{t1} \subseteq \operatorname{int}(F^t)$ with diameters $< 2^{-(n+1)}$ and $\tau(F^{t0} \cup F^{t1}) < 2^{-(n+1)}$. Set

$$N := \bigcap_{n \geq 0} \bigcup_{t \in \{0,1\}^n} F^t.$$

The shrinking-diameter compact-nested intersection gives a homeomorphism $2^\omega \to N$ by $t \mapsto \bigcap_n F^{t|_n}$; N is compact, zero-dimensional, has no isolated points, $\tau(N) = 0$. See [4, Thm. 6.2] for the general Cantor-scheme embedding of 2^ω into a nonempty perfect Polish space.

Hausdorff–Alexandroff surjection. Every compact metric space is a continuous image of 2^ω [4, Thm. 4.18]. Composing with the homeomorphism $N \cong 2^\omega$ yields a continuous (hence Borel) surjection $\psi : N \to C_0^\dagger$.

Modified labeling. Define

$$w^*(m) := \begin{cases} \psi(m) & m \in N, \\ w_0(m) & m \in M \setminus N. \end{cases}$$

w^* is Borel as a piecewise Borel function on the Borel partition $(N, M \setminus N)$, equals w_0 τ-a.e., and has image $w^*(M) = \psi(N) \cup w_0(M \setminus N) = C_0^\dagger \cup w_0(M \setminus N) = C_0^\dagger$ (because $w_0(M) \subseteq C_0^\dagger$ and ψ surjects). Hence $w^*(M) = C_0^\dagger$ is compact.

Profile-realization, global form. The profile-realization map $\Phi : \hat{\Sigma} \to W$ (continuous on the compact private-strategy space $\hat{\Sigma}$, with $W = \Phi(\hat{\Sigma})$ and nonempty compact fibers; see §2) admits a Borel right inverse $R : W \to \hat{\Sigma}$ by Kuratowski–Ryll-Nardzewski measurable selection [2, Thm. 18.13]. Define the modified agent strategy by the *global* composition

$$\hat{\sigma}^*(m) := R(w^*(m)), \qquad \sigma^*(m, \theta) := \hat{\sigma}^*(m)(\theta) \quad \text{for all } m \in M.$$

$\hat{\sigma}^*$ is Borel as the composition of Borel maps. Joint Borel measurability of $(m, \theta) \mapsto \sigma^*(m, \theta) = \hat{\sigma}^*(m)(\theta)$ follows because the evaluation map $\mathrm{ev} : \hat{\Sigma} \times \Theta \to \Delta(A)$, $(\hat{\sigma}, \theta) \mapsto \hat{\sigma}(\theta)$, is Borel on the admissible strategy space of §2. By construction, the induced payoff profile is $\Phi(R(w^*(m))) = w^*(m)$. The global form bypasses any seam-measurability concern at ∂N.

Value preservation. Since $w^* = w_0$ on $M \setminus N$ and $\tau(N) = 0$, the aligned integral $\alpha \int_M m \cdot w^*(m)\, \tau(dm) = \alpha \int_M m \cdot w_0(m)\, \tau(dm)$. The misaligned term depends on w only through the closure of its image (Lemma 1); $\overline{w^*(M)} = C_0^\dagger = \overline{w_0(M)}$, so the misaligned integral is unchanged. Hence $U(\sigma^*) = U(\sigma_0) = U^*$.

Exact contact for every s, with measurable selector inside N. For each $s \in M$, the rowwise infimum $r(s) = \min_{z \in C_0^\dagger} s \cdot z$ is attained on the compact $C_0^\dagger$. Consider the correspondence

$$H : M \rightrightarrows N, \qquad H(s) := \{m \in N : \psi(m) \in \arg\min_{z \in C_0^\dagger} s \cdot z\}.$$

H has nonempty compact values: nonempty because ψ surjects onto $C_0^\dagger$ and the rowwise minimum is attained; compact because $\{m \in N : \psi(m) \in K\}$ is the preimage under the continuous $\psi : N \to C_0^\dagger$ of a closed subset $K \subseteq C_0^\dagger$, hence closed in the compact N. H has Borel (in fact closed) graph because the defining condition $s\cdot\psi(m) = r(s)$ is continuous in (s, m). By Kuratowski–Ryll-Nardzewski [2, Thm. 18.13], H admits a Borel selector $m^* : M \to N$ with $m^*(s) \in H(s)$ for *every* s (not merely τ-a.e.). Set $\beta^*(\cdot \mid s) := \delta_{m^*(s)}$; the map $m \mapsto \delta_m$ is continuous from M into $\Delta(M)$ with the weak topology, so $\beta^* \in B$. For every s, $s \cdot w^*(m^*(s)) = s \cdot \psi(m^*(s)) = r(s)$. Therefore

$$\begin{aligned} U(\beta^*, \sigma^*) &= \alpha \int_M m \cdot w^*(m)\, \tau(dm) + (1-\alpha) \int_M s \cdot w^*(m^*(s))\, \tau(ds) \\ &= \alpha \int_M m \cdot w^*(m)\, \tau(dm) + (1-\alpha) \int_M r(s)\, \tau(ds) \;=\; F(C_0^\dagger) \;=\; U^* \end{aligned}$$

by Lemma 1. In particular $G(s) \neq \emptyset$ for every s, and exact-contact holds in the form of §6. □

Remark 10 (Atomlessness is a primitive canvas condition). Atomlessness of τ is a primitive structural hypothesis on the adviser-signal distribution. It is *not* logically comparable to exact-contact (which is an endogenous property of a chosen labeling): atomlessness is a sufficient *primitive* canvas condition under which a value-preserving null modification of any optimal labeling yields an exact-contact representative. The minimal technical add-on is the existence of a τ-null Borel $N \subseteq M$ admitting a Borel surjection onto $C_0^\dagger$; on compact metric standard Borel M, atomlessness is the cleanest primitive guaranteeing this.

Remark 11 (Atomless τ as economic primitive). The unconditional finite-M case in [1] relies on $|M| < \infty$ and $\alpha > 0$ to make every message q_β-on-path; this is at the opposite end of the distributional spectrum from atomless τ. In the infinite-M setting, atomlessness corresponds to continuously distributed adviser posteriors — the natural primitive when Θ is a continuum or when the adviser observes a continuous signal, including the binary-state continuous-density setting and the spherical example worked in [1]. Theorem 9 is therefore a clean continuous-signal add-on, not a hidden D&S primitive.

Remark 12 (Tier 2 is *not* addressed). Theorem 9 does not provide menu-Hall. The no-free-dust theorem (§10) rules out repairing a menu-engine calibration failure by routing q_β-mass through τ-null messages; that obstruction is orthogonal to exact-contact. Tier 2 still requires the menu-Hall calibration of §9 or its sufficient instantiations.

9 Tier 2: menu-Hall calibration

The menu-Hall condition asks that the rowwise minimization can be implemented by a kernel $\kappa(\cdot \mid s)$ whose induced posterior at the messages is Bayes-rationalizable for the agent's continuation $\hat{\sigma}^*(m)$. Two equivalent forms:

- **Kernel form.** $\kappa(G(s) \mid s) = 1$ τ-a.e., and $P_{\gamma_\alpha}(\cdot \mid m) \in B(m)$ q-a.e. where $\gamma_\alpha := \alpha(\mathrm{id}, \mathrm{id})_{\#}\tau + (1-\alpha)\tau \otimes \kappa$ and $q := (\gamma_\alpha)_2$.
- **Support-function form.** For every measurable $E \subseteq M$ and every continuous affine $\phi : \Delta(\Omega) \to \mathbb{R}$,

$$\alpha \int_E \phi(m)\, \tau(dm) + (1-\alpha) \int_M \int_E \phi(s)\, \kappa(dm \mid s)\, \tau(ds) \;\leq\; \int_E h_{B(m)}(\phi)\, q(dm),$$

  where $h_{B(m)}(\phi) := \sup_{\mu \in B(m)} \phi(\mu)$.

The support-function inequality, taken over a countable separating family of affine ϕ and all measurable E, is equivalent to the pointwise membership statement $P_{\gamma_\alpha}(\cdot \mid m) \in B(m)$ q-a.e.; equivalence uses finite-dimensional separation in $\Delta(\Omega)$.

This is exactly the per-message Bayes-optimality requirement of Definition 2. Tier 2 follows.

10 Sharpness of menu-Hall

Menu-Hall is genuinely needed inside the engine. The witness is ternary winner-takes-all: $\Omega = \{0, 1, 2\}$, $A = \{a_0, a_1, a_2\}$ with $u(a_\omega, \omega) = 1$ and $u(a_i, \omega) = -1$ for $i \neq \omega$, prior $\mu_0 = (1/3, 1/3, 1/3)$, atomless full-support τ on $\Delta(\Omega)$, and trust region $T := \{\mu : \mu(0) \leq 0.4\}$. The induced payoff-profile menu under any plurality continuation is the full vertex set $C^\dagger = \{v_0, v_1, v_2\}$ with $v_\omega(\omega) = 1$ and $v_\omega(\omega') = -1$ for $\omega' \neq \omega$.

For each nonempty $I \subseteq \{0, 1, 2\}$, define the rowwise minimizer cone and Bayes-optimality cone of the mixed profile $w_\lambda := \sum_{i \in I} \lambda_i v_i$ (with $\operatorname{supp} \lambda = I$):

$$K_I^- \; := \; \{s \in \Delta(\Omega) : s_i \leq s_k \; \forall i \in I, \forall k \in \{0, 1, 2\}\},$$

$$B_I \; := \; \{p \in \Delta(\Omega) : p_i \geq p_k \; \forall i \in I, \forall k \in \{0, 1, 2\}\}.$$

(K_I^- uses the identity $s \cdot w_\lambda = 2 \sum_i \lambda_i s_i - 1$ to identify rowwise minimizers; B_I similarly identifies the labels making w_λ Bayes-optimal at posterior p.)

Lemma 13 (Cone intersection)**.** *For every nonempty $I \subseteq \{0, 1, 2\}$, if ρ is a Borel probability on $\Delta(\Omega)$ with $\rho(K_I^-) = 1$ and barycenter $\bar{s} \in B_I$, then $\rho = \delta_{\mu_0}$ where $\mu_0 = (1/3, 1/3, 1/3)$.*

Sketch. Fix $i \in I$. ρ-a.s., $s_k - s_i \geq 0$ for every k. Since $\bar{s} \in B_I$, $\bar{s}_i \geq \bar{s}_k$, so $\int (s_k - s_i)\, d\rho \leq 0$. A bounded nonnegative Borel random variable with nonpositive expectation is zero a.s., hence $s_k = s_i$ ρ-a.s. for every k. Coordinates summing to one force $s = (1/3, 1/3, 1/3)$ ρ-a.s. □

Theorem 14 (No-free-dust)**.** *Under atomless τ in the ternary winner-takes-all setting, no Borel τ-null set $N \subseteq M$, no Borel labeling $w_N : N \to W$, and no adversarial kernel κ supported on rowwise minimizers can simultaneously satisfy: $q_\beta(N) > 0$ where $q_\beta = \alpha\tau + (1 - \alpha)(\tau \otimes \kappa)_2$, AND Bayes-cone calibration q_N-a.e. on N (where q_N is the dust message marginal).*

Sketch. Define $\nu(ds, dm) := \tau(ds)\, \kappa(dm \mid s)$ on $\Delta(\Omega) \times M$, and let $\nu_N := \nu \restriction_{\Delta(\Omega) \times N}$. Disintegrate ν_N over its second marginal q_N: $\nu_N(A \times E) = \int_E \rho_m(A)\, q_N(dm)$ for Borel kernels ρ_m, q_N-a.e. $m \in N$. The rowwise-minimizer support gives $\rho_m(K_{I(m)}^-) = 1$ q_N-a.e. The Bayes-cone calibration gives the barycenter of ρ_m in $B_{I(m)}$. By Lemma 13, $\rho_m = \delta_{\mu_0}$ q_N-a.e. Hence

$$\nu(\{\mu_0\} \times N) \; = \; \int_N \rho_m(\{\mu_0\})\, q_N(dm) \; = \; q_N(N).$$

Now $\tau(N) = 0$, so $q_\beta(N) = (1 - \alpha)\, q_N(N)$; assumption $q_\beta(N) > 0$ gives $q_N(N) > 0$ (assuming $\alpha < 1$; $\alpha = 1$ rules out adversarial dust trivially). On the other hand, ν has first marginal τ, so $\nu(\{\mu_0\} \times N) \leq \nu(\{\mu_0\} \times \Delta(\Omega)) = \tau(\{\mu_0\}) = 0$ (atomlessness). Contradiction. □

What this buys. Lemma 13 strengthens the sharpness from one boundary point ($I = \{0\}$, $t_0 = (0.4, 0.3, 0.3)$, recovering $K_0^- \cap B_0 = \{\mu_0\}$) to a uniform statement covering all support patterns. Theorem 14 then closes the most plausible escape: an infinite-setting reading of Definition 2 (q_β-a.e., §1) explicitly allows adversaries to use τ-null messages, and one might hope to repair calibration by routing extra labels through such "null-message dust." The proof never counts messages: it works for finite, countable, or continuum dust. The obstruction is invariant to deterministic vs. mixed kernels, pure vs. mixed dust labels, and atomic vs. diffuse dust.

Sharpness for the boundary witness. Setting $I = \{0\}$ in Lemma 13 recovers the pointwise sharpness at $t_0 = (0.4, 0.3, 0.3)$: the source cone is $K_0^- = \{s : s_0 \le s_1, s_0 \le s_2\}$, the Bayes cone for action a_0 is $B(t_0) = \{p : p_0 \ge p_1, p_0 \ge p_2\}$, and $K_0^- \cap B(t_0) = \{\mu_0\}$. With atomless τ, no positive source mass concentrates at μ_0. Theorem 14 then shows no null-message dust construction can repair this. menu-Hall is therefore genuinely required for Tier 2 in this geometry.

11 Classification of the witness

Lemma 13 and Theorem 14 establish that menu-Hall is needed inside the menu engine. They do *not* falsify unrestricted Theorem 2.

The trust region T used in §10 is not primitive. Concretely, $T = \{\mu : \mu(0) \le 0.4\}$ contains beliefs with each plurality label: $(0.4, 0.3, 0.3) \mapsto a_0$, $(0.1, 0.8, 0.1) \mapsto a_1$, $(0.1, 0.1, 0.8) \mapsto a_2$. Any plurality-vertex continuation on T produces all three pure profiles v_0, v_1, v_2, so the induced effective menu is the full vertex menu. The off-T Bregman/TRS projection chooses an interior point of T whose induced profile maximizes $m \cdot w$ over the in-T menu, but since the in-T menu is already the full vertex set, the projection collapses to ordinary plurality at m. Thus every $m \in \Delta(\Omega)$, inside or outside T, induces the same Bayes-optimal WTA vertex, identically to $T = \Delta(\Omega)$.

Consequence. If the full menu $\{v_0, v_1, v_2\}$ is optimal under some primitive $(\alpha, \tau, \Theta, f, u, A)$, then the same behavior is representable as $T = \Delta(\Omega)$. If the full menu is not optimal, neither is the halfspace. Either way, the boundary number 0.4 and the boundary point t_0 are *representational scenery, not load-bearing beams.* The witness demonstrates menu-Hall is genuinely needed inside the F-functional optimization (i.e., choosing among compact menus in $\mathcal{K}(W)$), but it does *not* certify that every primitive optimal solution must hit the same obstruction. To recover the obstruction, the strategy would need to label a τ-null boundary point by v_0 while sourcing it from K_0^-, but the same trust region already contains all three vertices, so the primitive induced menu is full.

12 Comparison to the earlier route

The earlier route worked in the agent's strategy space with the Balder weak topology, used Mertens's asymmetric minmax theorem on a τ-dominated restricted game, and lifted the restricted maximizer back to the unrestricted game by a Lusin regularization. It required three additional hypotheses:

- **(A5-thick)** an endogenous Lusin-thickness condition on the maximizer's representative;

- **(A8c-attain)** attainment of the rowwise infimum at a measurable selector;
- **(TRE-gen-Hall)** a deterministic version of the present menu-Hall calibration.

The comparison to the current route is summarised below.

Conclusion	**Earlier route**	**Current route**
σ^* optimal	standing + A5-thick	standing alone
ε-adversary	standing + A5-thick	standing alone
exact β^*	standing + A5-thick + A8c-attain	standing + exact-contact
robust rat'lity	ditto + TRE-gen-Hall (deterministic)	ditto + menu-Hall (set-valued)
Engine	Balder weak topology on Σ, Mertens minmax, Lusin lift	compactness of $\mathcal{K}(W)$ in finite-dim $\mathbb{R}^{\|\Omega\|}$
Sharpness	pointwise ternary witness	cone intersection (uniform) + no-free-dust theorem
Witness	possible Th. 2 obstruction (then unclassified)	menu-engine artefact, classified not a Th. 2 counterexample

The architectural shift is moving from infinite-dimensional strategy compactness to finite-dimensional payoff-profile compactness. Four substantive consequences:

1. Tier 1a is now *unconditional*. A5-thick (a Lusin-thickness clause that fails in perfect-revelation examples) and A8c-attain (a rowwise-argmin attainment clause that fails on upward-spike rows) are both eliminated for value optimality and ε-adversaries.

2. Exact-contact is endogenous to the optimal labeling, so it can be checked from the chosen w^* rather than from a representative of an abstract maximizer. The sufficient routes (closed image, closed-graph correspondence, u.h.c. Bayes-action) are all instantiable by primitive choices.

3. Menu-Hall is set-valued, and the support-function form makes the calibration condition a transparent integrated separation inequality. This is strictly weaker than deterministic TRE-gen-Hall (every deterministic m^* kernel is a degenerate κ).

4. The sharpness package is uniform: Lemma 13 covers all support patterns, and Theorem 14 closes null-message dust escapes that the q_β-a.e. reading of Definition 2 (§1) explicitly allows. The witness's trust region is classified (§11) as a menu-engine artefact, not a counterexample to unrestricted Theorem 2.

References

[1] Piotr Dworczak and Alex Smolin. *Robust Trust*. arXiv:2602.09490, 2026.

[2] Charalambos D. Aliprantis and Kim C. Border. *Infinite Dimensional Analysis: A Hitchhiker's Guide*. Springer, 3rd edition, 2006. (Specifically: Theorems 18.13 (Kuratowski–Ryll-Nardzewski / Jankov–von Neumann), 18.19 (measurable maximum), 15.11 (compactness of Δ on compact metric).)

[3] Vladimir I. Bogachev. *Measure Theory*. Springer, 2007. (Specifically: Theorem 7.1.7 (Vol. II): inner regularity of Borel probability measures on Polish spaces; standard-Borel disintegration; Polish-valued measurable selection.)

[4] Alexander S. Kechris. *Classical Descriptive Set Theory*. Graduate Texts in Mathematics 156, Springer, 1995. (Specifically: Theorem 4.18 (Hausdorff–Alexandroff: every compact metric space is a continuous image of the Cantor space 2^ω); Theorem 6.2 (Cantor-scheme embedding of 2^ω into a nonempty perfect Polish space).) (Standard-Borel disintegration; Polish-valued measurable selection.)